

Assignment Draft: Building a Social Media Backend in Go

Objective

Develop a **RESTful API** using Go that mimics a simple social media application where users can create and browse posts. The application must support **user authentication with JWT**, browsing all posts, and searching posts based on user ID.

Core Requirements

1. Authentication

- Implement user registration and login.
- Use **JWT (JSON Web Tokens)** for authenticating and authorizing users.
- Users must be authenticated via JWT to access post-related endpoints.

2. Posts Functionality

- **Create Post:** Authenticated users can create new posts.
- **Browse All Posts:** Any authenticated user can fetch all posts from all users.
- **Search Posts:**
 - By **User ID:** Retrieve posts made by a specific user based on their unique ID.

3. API Design

- All endpoints must follow RESTful conventions.
- Example endpoints:
 - POST /register — User registration.
 - POST /login — User login (returns JWT).
 - POST /posts — Create a new post (JWT required).
 - GET /posts — Get all posts (JWT required).
 - GET /posts?user_id={id} — Get all posts by a particular user (JWT required).

4. Architecture Pattern

Your Go application must be organized with the **handler-service-repository** pattern:

- **Handler:** Handles HTTP requests and responses.
- **Service:** Contains business logic for authentication and post management.
- **Repository:** Acts as a data layer, interacting with the database or in-memory storage.